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Overview

This project aims at developing a machine learning based credit risk model to predict loan default probability using both customer financial profiles and transaction behaviour data. This study integrates demographic, income, and loan-related variables with behavioural spending indicators to create a more comprehensive risk assessment framework.

The model applies

Data Preprocessing

Feature Engineering

Classification Techniques

including Random Forest algorithms and cross validation methods, to improve prediction accuracy and reliability. The study also groups customers based on their financial behaviour and uses data analysis to understand how spending patterns and home ownership affect the chances of loan default.

FLAG 1

Transaction-Augmented Credit Risk Prediction

Objective:

This project aims to improve loan default prediction by augmenting traditional borrower and loan attributes with transaction-level behavioral features, testing whether observed spending patterns provide additional predictive power over standard credit risk variables.

Dataset:

Two datasets are used:

- A loan application dataset (58,645 observations) containing borrower demographics, loan characteristics, credit history, and a binary loan default indicator.
- A transaction dataset (101,915 records), aggregated at the user level to derive behavioral metrics such as total spend, transaction frequency, spend volatility, geographic diversity, and primary transaction mode.

The merged dataset contains 58,645 borrowers with 20 features, ensuring one behavioral profile per individual.

Methodology:

Transaction data is aggregated and merged with loan-level information. Supervised classification models are trained to predict default, and feature importance analysis is used to assess the incremental value of transaction-derived variables relative to conventional credit indicators.

Evaluation Strategy:

Model performance is evaluated using ROC-AUC, precision, recall, and F1-score, with out-of-sample validation. Results are compared between baseline models and transaction-enriched

models to quantify performance improvements.

Expected Impact:

The study supports behavior-aware credit assessment, enabling lenders to better identify hidden risk, improve decisions for thin-file borrowers, and reduce default rates through more data-driven lending strategies.

FLAG 2

Data Cleaning & Model Readiness (EDA Summary)

Dataset Structure

- Final dataset contains 58,645 observations and 20 features, created by merging loan application data with aggregated user-level transaction data.
- Each record represents a unique borrower, suitable for supervised classification.

Variable Types and Feature Composition

The data includes both **numerical variables**—such as borrower demographics, loan attributes, affordability ratios, and transaction-based behavioral indicators—and **categorical variables** capturing loan intent, loan grade, home ownership, credit history flags, and transaction mode. The target variable, `loan_status`, is binary, indicating repayment or default.

Data Quality and Distributional Characteristics

No material missing values are observed after merging. Several numerical features, particularly **loan-to-income ratio and spend volatility**, display right-skewed distributions with upper-tail outliers. These values are economically meaningful indicators of financial stress and are therefore retained. The target variable exhibits **significant class imbalance**, with roughly **86% non-defaults and 14% defaults**.

Key Risk Relationships and Visual Insights

Exploratory analysis reveals a strong, monotonic increase in default probability from **loan grade A to G**, validating loan grade as a primary risk signal. Defaulters show higher and more dispersed loan-to-income ratios, indicating affordability pressure, and are concentrated at higher interest rates, reflecting risk-based pricing. Visual diagnostics confirm these patterns through class distribution, grade-wise default, leverage dispersion, and interest rate density plots.

Model Readiness

Overall, the dataset is **clean, economically consistent, and model-ready**, with clear risk separation, meaningful behavioral features, and structural characteristics that support robust and interpretable credit risk modeling.

FLAG 3

To create a baseline, we commenced with a Random Forest Classifier. The model was set up with 100 trees (`n_estimators = 100`) and `class_weight` was set to “balanced” to account for the imbalance in default cases. Random Forest has been selected as it is relatively resistant to overfitting and shows a good performance with non linear relationships among features such as income and loan grade, without requiring heavy feature engineering.

For Training and Validation Setup, firstly we have splitted the data into 80:20 train test models. From our result we have come to the conclusion that only 14% of the borrowers default. So, to deal with this we have applied stratified sampling (stratify = y) to make sure that both training and testing datasets maintained the same class distribution. This has helped us in avoiding biased evaluation caused by under- representation of defaulters.

For Evaluation Metrics, the accuracy alone was not sufficient given the class imbalance. So, we focussed on ROC-AUC, to measure how well the model distinguishes the defaulters from non defaulters overall. The F1 score for the default class (Class 1), to balance precision & recall and specifically evaluate the performance on default prediction.

The Baseline Results are as follows : Accuracy = 95.11% , ROC-AUC = 0.9323 , F1 Score = 0.8052

Coming to the interpretation, the baseline model performs strongly, with an ROC-AUC of 0.93, signifying the static application variables such as loan_grade and loan_percent_income that already contain meaningful credit risk signals.

The precision is high (0.93), which is noticeably lower than (0.71). This means the model still misses around 29% of actual defaulters. This highlights a limitation of completely relying on application data and motivates the next step (Flag 4), where transactional features have been found to improve the model's ability to detect these missed default cases.

FLAG 4

After the Transactional Information Gain, we incorporated it into the static data, so now we have trained two identical Random Forest Models to understand if the transaction data actually improves predictions.

For the Baseline Model, we have used only 11 basic application features such as person_income, loan_grade, and loan_percent_income.

For the Enhanced Model, we have used 20 features with an addition to the transaction based variables like total_spend, spend_volatility, and transaction_intensity.

Below is the comparison of the performance:

The accuracy for the Baseline Model was 95.11%, and for the Enhanced Model was 94.94% with a change of -0.17%. The F1 Score for the Baseline Model is 0.8040, and that for Enhanced Model is 0.7943 with a change of -0.0096. The ROC-AUC for the Baseline Model is 0.9304, and that for the Enhanced model is 0.9305 with a change of 0.0001.

Adding the transaction data did not improve the model much. In fact the performance dropped slightly due to the reasons listed below :

Firstly, there were too many users who had very little transaction history which means there was scarcity of transaction records for many users. So, for them we had to fill the missing values with either zero or averages. The values which were filled added noise to the data and weakened the strong signals which were already coming from application features.

Secondly, the application data is already very powerful. The Baseline model performed really well (ROC-AUC approximately around 0.93). Features like `loan_grade` and `loan_percent_income` turned out to be the strongest predictors of default. Simply, it showcases how affordable the loan is compared to income matters more than daily spending behaviour.

Thirdly, we discovered a high correlation between `total_spend` and `person_income`, so the transaction data was effectively working as a proxy for income. We found that `total_spend` is related to `person_income`. So, instead of giving new information, transaction features which were mostly repeated what income already told us. Therefore, it provided validation rather than new information gain.

With this we came to the conclusion that though transaction data adds behavioural details yet it does not currently improve the prediction accuracy what basic application data already provides. The model is capturing most of the default risk utilizing demographic and loan related features alone.

FLAG 5

Behavioral Statistic: Transaction Volatility Coefficient (TVC)

We have created a new metric named the Transaction Volatility Coefficient (TVC) to measure how unsteady a person's spending behavior is compared to their usual transaction size. It is computed by dividing a user's spending volatility (standard deviation) by their average transaction amount.

Evidence of Behavioral Differences

The results showed a meaningful behavioural gap between defaulters and non defaulters. Defaulters have a positive average TVC of 1.79 which signifies a typical spending pattern where

higher volatility goes conjointly with higher consumption. Whereas, non defaulters on the other hand have a negative average TVC of -8.64.

This showcases a 120.76% shift and most importantly, there is a reversal in sign. This shows us that safe borrowers and risky borrowers are not only different in magnitude but also they are behaving in different ways fundamentally.

It actually means, this polarity shift reveals a deep difference in how the two groups are managing money. Talking about non defaulters, the negative TVC suggests their transaction histories are dominated by inflows such as salary credits, refunds, or other deposits. This pattern brings in limelight the stronger liquidity and financial stability. Whereas, for defaulters, the positive TVC signifies a “net spender” profile, where money is consistently flowing out of the account creating continuing financial pressure without sufficient incoming funds.

With this we arrived at the conclusion that, these findings show that high risk borrowers do not simply spend more or less. They just operate with a completely different financial direction. Safe borrowers tend to have net inflows while risky borrowers experience net outflows.

FLAG 6

Flag 6: Temporal Awareness

1. Statistic Submitted Intraday Spending Velocity (ISV)

2. Data Sources & Methodology

Source Files: We joined transactions_datathon_ready.csv (Transaction Log) with train.csv (Target Labels) using the user_id key.

Columns Analyzed: date (Timestamp), amount (Transaction Value), and loan_status (Target).

Calculation: We parsed the date column to bucket transactions by Hour (0–14). ISV is defined as the percentage drift in the mean amount from the first hour (Hour 0) to the last observed hour (Hour 14).

3. Quantitative Evidence The time-series analysis of the date column reveals a distinct divergence in spending momentum between the two loan_status groups:

Non-Defaulters (loan_status = 0): The average amount increased by +147.83% over the observed window.

Defaulters (loan_status = 1): The average amount increased by +157.19% over the same window.

4. Interpretation The analysis identifies a specific Velocity Risk Pattern. Defaulters accelerate their spending 9.36% faster than non-defaulters as the day progresses. This finding confirms that the slope of the amount over time (derived from date) is a predictive feature orthogonal to static aggregates like total_spend, capturing a "Bust-Out" dynamic where high-risk users exhibit steeper spending acceleration.

FLAG 7

Segment Sensitivity Analysis

Firstly, the segment identified the Housing Status (person_home_ownership) from dataset train_with_transactions.csv.

Secondly, there is a comparison of Subgroups where we have analyzed the outcome distribution and behavioral patterns across two distinct segments to assess model sensitivity where

Segment A (with High Risk) are Renters (with a count of 30,594) and Segment B (with Low Risk) are Owners (with a count of 3,138).

Thirdly, coming to the quantitative evidence, the analysis reveals a 1,524 divergence in risk outcomes in spite of nearly identical spending behaviour.

Shedding some light upon the outcome sensitivity (risk), the renters default rate is 22.26%, the owners default rate is 1.37%. Talking about the sensitivity, the renters are 16.2x more likely to default than owners (a + 1,524.17% risk multiplier).

Talking about the Behavioural sensitivity (Volatility), the renters average volatility is 77.81, the owners average volatility is 80.03. The difference in spending stability is negligible (-2.77%).

The analysis demonstrates the extreme outcome sensitivity to asset ownership. The renters and owners exhibit a -2.8% difference in daily spending variance (specifying similar financial behaviour), where the risk profiles of both are radically different. This proves that `person_home_ownership` is a critical discriminator. The default risk is driven by asset backing (wealth) rather than the spending stability (behaviour) for this specific segmentation.

FLAG 8

Flag 8: Model Robustness & Stability Analysis

1. Stability Evaluation (Cross-Validation)

To evaluate the model's reliability under varying data splits, a **5-Fold Stratified Cross-Validation** was performed on the training set.

Results:

- **Mean ROC-AUC:** 0.9247
- **Standard Deviation:** 0.0042 ($\pm 0.42\%$)

Conclusion:

The extremely low variance (Standard Deviation < 0.01) indicates that the model is highly stable and does not rely on specific data partitions to achieve its performance.

2. Robustness Stress Test (Noise Injection)

The model's resilience was tested by injecting **20% Gaussian noise** into the primary feature (`person_income`) to simulate data quality degradation.

Results:

- **Original AUC:** 0.9247
- **Stressed AUC:** 0.9177
- **Performance Drop:** -0.0070 (-0.76%)

Conclusion:

The model demonstrated high robustness, retaining **99.2%** of its predictive power despite significant feature corruption. This confirms that the decision boundaries are well-generalized and resilient to input volatility.

FLAG 9

1. Insight Title

Interaction Effect: Housing Status as a Moderator of Debt-to-Income Risk

2. Analytical Finding

The analysis identifies a significant **interaction effect** between **person_home_ownership** and **loan_percent_income (DTI)**. While DTI is commonly modeled as a linear risk factor, the results indicate that its predictive power varies depending on asset ownership.

High debt burdens (>0.30) are strongly predictive of default among **Renters**, but statistically insignificant among **Homeowners**.

3. Quantitative Evidence (Stratified Analysis)

Result A: DTI Risk Divergence

Borrowers with **High DTI (>0.30)** were segmented to isolate the effect of housing status.

- **Renters (High DTI):**
 - Default Rate = 95.7% (n = 3,219)

- **Owners (High DTI):**
 - Default Rate = 4.9% (n = 344)

Delta:

Asset ownership reduces the default probability of high-leverage borrowers by **90.8 percentage points** (Risk Ratio: 19.4).

Result B: Grade Misclassification

“Prime” and “Sub-Prime” borrower segments were compared while controlling for housing status.

- **Grade A Renters:**
 - Actual Default Rate = 10.66% (n = 8,244)
- **Grade B Owners:**
 - Actual Default Rate = 0.56% (n = 1,065)

Implication:

The existing credit scoring model underestimates default risk for Renters by approximately **18.9 times** relative to Owners in adjacent loan grades.

4. Statistical Validation (Multivariate Logistic Regression)

To control for potential confounding variables such as **Income, Age, and Employment Length**, a Logistic Regression model was estimated using a sample size of **N = 58,645** (Pseudo R^2 = 0.229).

Key Results:

- **Adjusted Odds Ratio (Rent vs Own):** 30.49 ($p < 0.001$)

Interpretation:

Renters are approximately **30.5 times** more likely to default than homeowners, after controlling for income and DTI.

Robustness Check:

- **95% Confidence Interval for High-DTI Owners:** [2.65%, 7.23%]

Conclusion:

The confidence interval does not overlap with the renter default rate (95.7%), confirming that the housing ownership effect is statistically significant and robust.

5. Recommendation

Revise the credit policy to incorporate a **Housing-Adjusted DTI Threshold**.

For Renters:

- Implement a strict DTI cap at **0.30**, as default risk exceeds 95% beyond this threshold.

For Homeowners:

- Relax the DTI cap to **0.45**, where default risk remains below the upper confidence interval of 7.3%.

Expected Impact:

This adjustment is expected to reduce overall portfolio loss rates by eliminating high-risk renter exposures while retaining low-risk, high-DTI homeowner lending opportunities.

Firstly, the interaction effect is given as housing status as a moderator of debt to income risk.

Secondly, the analysis showcases a significant interaction effect between `person_home_ownership` and `loan_percent_income` (DTI). The DTI is modelled as a linear risk factor where the results indicate that its predictive power differs depending on the asset ownership. High debt burdens (with >0.30) are strongly predictive of default among the renters, but it is statistically insignificant among homeowners.

The quantitative evidence (stratified analysis) for this is given as:

The Result A: DTI Risk divergence

Borrowers with high DTI (>0.30), were segmented to isolate the effect of housing status. The default rate for renters (with high DTI) is 95.7% ($n=3219$) and the default rate for owners with high DTI is 4.9% ($n=344$). Asset ownership lowers the default probability of high-leverage borrowers by 90.8%, corresponding to a risk ratio of 19.4.

The Result B: Grade Misclassification

Comparing 'Prime' and 'Sub-Prime' borrower segments while controlling for housing status, the Grade A renters shows an actual default rate of 10.66% (n = 8,244) compared to just 0.56% among the Grade B owners (n = 1,065), denoting that the existing credit scoring model underestimates default risk for renters by approximately 18.9x relative to the owners in adjacent loan grades.

Multivariate logistic regression is used to control for income, age, and employment length (N = 58,645; Pseudo R^2 = 0.229), renters proved to be approximately 30.5x more likely to default than homeowners (adjusted odds ratio = 30.49, $p < 0.001$), and a robustness check denoted a 95% confidence interval of [2.65%, 7.23%] for high-DTI homeowners that does not intersect with the renter default rate of 95.7% confirming that the effect of housing ownership on default risk is statistically significant and robust.

The recommendation is to revise the credit policy by introducing a housing adjusted DTI threshold, with a strict cap of 0.30 for renters whose default risk exceeds 95% beyond this level and a relaxed cap of 0.45 for homeowners, where default risk remained below 7.3%, a change expected to decrease overall portfolio losses while maintaining low risk lending opportunities.